

GreenSure

Software Requirements Specification

Carbon Footprint Tracking & Green Insurance Platform

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the GreenSure platform. It is intended to serve as a comprehensive reference for development, testing, and future enhancements of the system.

1.2 Project Overview

GreenSure is a full-stack web application that calculates a user's annual carbon footprint and links that score directly to an insurance premium discount. The core motivation is to address the global carbon emissions problem by giving individuals and businesses a direct financial incentive to live and operate more sustainably.

"The greener you live, the less you pay."

GreenSure is not just an insurance application. It is a behaviour change platform that uses money as the motivator to push users toward a greener lifestyle year on year.

1.3 Scope

The GreenSure platform covers the following scope:

- Annual carbon declaration submission by households and MSMEs
- Field agent physical verification of declared data with GPS proof
- Automated carbon score calculation using the CarbonScoreService engine
- Zone-based insurance premium discount allocation
- Year-on-year improvement bonus for users who reduce their carbon score
- Role-based access control across four roles: User, Field Agent, Admin, and System
- Automated email notifications for all key events
- Three Spring Schedulers running automated background processes

1.4 Definitions & Acronyms

Term / Acronym	Definition
SRS	Software Requirements Specification
SPA	Single Page Application — Angular frontend loaded once, routes handled client-side
RBAC	Role-Based Access Control — access enforced based on user role
JWT	JSON Web Token — stateless authentication token
MSME	Micro, Small and Medium Enterprise — a business user type in GreenSure
HH	Household — a residential user type in GreenSure
CO ₂	Carbon Dioxide — unit of carbon emission measurement
CarbonScore	The verified annual CO ₂ score assigned to a user after agent verification
Zone	Green Champion, Improver, or Defaulter — classification based on carbon score
Scheduler	Spring @Scheduled background task running on fixed intervals
SMTP	Simple Mail Transfer Protocol — used for all outgoing emails

2. Overall Description

2.1 The Problem Being Solved

Global carbon emissions are rising every year. A significant contribution comes from individual and business lifestyle choices — energy consumption, vehicles driven, and operational waste. Despite widespread awareness, there is no direct financial motivation for individuals to change their behaviour.

Insurance companies have historically priced premiums based solely on postcode and age, completely ignoring a customer's environmental behaviour. A household with rooftop solar and an EV pays the exact same premium as a household running diesel generators and three petrol cars. There is no reward for making greener choices and no transparency on how premiums are calculated.

Key Market Gap:

73% of insurance customers want personalised premiums.

Less than 8% of insurers use behavioural data in pricing.

Companies globally are committing to net zero emissions targets —
yet individual insurance products do nothing to accelerate that goal.

2.2 Product Perspective

GreenSure fills this gap by building a platform where a user's verified carbon footprint is the primary driver of their insurance pricing. The platform creates a habit loop — every year a user sees their score, sees their discount, and sees exactly what they need to change to save more money next year. This loop is what drives real, measurable lifestyle change.

2.3 User Classes and Characteristics

2.3.1 User (Household / MSME)

Regular citizens or small business owners who submit their annual carbon declaration, receive a verified carbon score, and get insurance premium discounts based on that score. Users can be of two subtypes:

- Household — individual residential users tracking personal carbon footprint
- MSME — small business owners tracking business operational emissions

2.3.2 Field Agent

On-ground employees who physically visit homes and businesses to verify the carbon data submitted by users. Agents are the credibility layer of GreenSure — agent-verified data always overrides user-declared values.

2.3.3 Admin

Platform administrators who manage users, agents, declarations, assignments, and overall platform health. Admins have full visibility and control over all platform data.

2.3.4 System (Schedulers)

Three automated Spring Schedulers that run continuously in the background without any human trigger:

- AssignmentEngine — runs every 30 minutes, auto-assigns agents to unassigned declarations
- DeadlineMonitor — runs every hour, detects missed deadlines and reassigns automatically
- RenewalEngine — runs on the 1st of every month at 9AM, sends renewal reminder emails

2.4 Operating Environment

Backend:	Spring Boot 4.0, Java 21, Spring Security, JWT, Spring Data JPA, Hibernate, H2 (dev) / PostgreSQL (production), JavaMailSender / SMTP
Frontend:	Angular 21, TypeScript, Tailwind CSS, Lucide Icons, Angular Signals, RxJS, Auth Guards, HTTP Interceptors
Auth:	Stateless JWT — no sessions on server. Token carries userId, email, role, userType. Validated by JwtAuthFilter on every request.
Deployment:	Backend: AWS / Railway. Frontend: Vercel / Netlify.

3. System Architecture

3.1 Layered Architecture

GreenSure follows a 4-layer N-Tier architecture with strict separation of concerns between each layer:

Layer	Technology	Responsibility
Layer 1 — Frontend	Angular 21, TypeScript, Tailwind CSS	SPA, role-based views, Auth Guards, HTTP Interceptors, user interaction
Layer 2 — Controllers	Spring Boot, Spring Security, @PreAuthorize	REST endpoints, JWT validation, RBAC enforcement, request/response handling
Layer 3 — Services	Spring @Service, Spring Scheduler, JavaMailSender	Business logic, carbon score engine, schedulers, email dispatch
Layer 4 — Data	Spring Data JPA, Hibernate, H2 / PostgreSQL	Entities, repositories, persistence, ORM mappings

3.2 Security Architecture

GreenSure implements stateless JWT-based authentication with backend RBAC enforcement:

- Every request passes through JwtAuthFilter before reaching any controller
- JWT token carries: userId, email, role (USER / AGENT / ADMIN), userType (HOUSEHOLD / MSME)
- @PreAuthorize annotations enforce role-based access at the method level
- Frontend Auth Guards and HTTP Interceptors handle UX-level RBAC and auto-attach tokens
- Frontend RBAC controls what the user sees. Backend RBAC controls what they can access. Both are independent.

3.3 Frontend Structure

The Angular frontend is organized under src/app/ as follows:

```
src/app/  
  features/  
    auth/      login, register, forgot-password, reset-password  
    user/      dashboard, profile, policies, my-policies, score-history,  
               recommendations, notifications, welcome  
    declaration/ declaration-start, declaration-fill, declaration-vehicles,  
               declaration-review, declaration-history  
    agent/     dashboard, task-detail, verify, performance, notifications  
    admin/     dashboard, users, user-detail, agents, create-agent,  
               agent-detail, declarations, assignments
```

shared/

components/ navbar, sidebar, alerts, badges, forms, inputs, charts

core/

guards/ AuthGuard, RoleGuard

interceptors/ JwtInterceptor, ErrorInterceptor

services/ AuthService and all feature services

4. Functional Requirements

4.1 Authentication Module

Req ID	Description	Priority	Role
FR-AUTH-01	User registers with full name, email, password, user type (HOUSEHOLD or MSME), and profile details. System creates user account and sends welcome email.	High	All
FR-AUTH-02	User logs in with email and password. System validates credentials, generates JWT token, and returns token with user details.	High	All
FR-AUTH-03	JWT token is validated on every request by JwtAuthFilter before any controller is reached.	High	System
FR-AUTH-04	User submits their email to the forgot password form. System generates a secure UUID token, saves it with 15-minute expiry, and sends a reset link to the email.	High	User
FR-AUTH-05	User clicks reset link, lands on reset password page with token in URL. System validates token and expiry. User sets a new password. System clears the token.	High	User
FR-AUTH-06	Login and registration forms show red border on invalid fields and auto-scroll to the first error on submit.	Medium	User

4.2 Carbon Declaration Module

Req ID	Description	Priority	Role
FR-DEC-01	User submits an annual carbon declaration through a multi-step form capturing energy usage (kWh), transport km, vehicle details, and lifestyle / operations data. Household and MSME forms have different fields.	High	User
FR-DEC-02	System enforces one declaration per user per year. Duplicate submissions are rejected with a custom exception.	High	System
FR-DEC-03	Declaration status follows this state machine: DRAFT → SUBMITTED → UNDER_VERIFICATION → VERIFIED. If rejected by agent: REJECTED.	High	System
FR-DEC-04	If declaration is in DRAFT status, user can edit and save it. If SUBMITTED, UNDER_VERIFICATION, or VERIFIED, the form is read-only.	High	User
FR-DEC-05	If declaration is REJECTED, user can edit and resubmit. Maximum 3 resubmissions allowed. System tracks resubmission_count. On exceeding limit, throws MaxResubmissionLimitExceededException.	High	User
FR-DEC-06	Declaration history page shows all past declarations. Clicking View Details shows details inline or in a modal — no navigation to a separate page.	Medium	User
FR-DEC-07	User can add, edit, and remove vehicles from their declaration. Vehicles are stored as a list linked to the declaration.	High	User

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Req ID	Description	Priority	Role
FR-DEC-08	On successful submission, system sends email confirmation to the user.	Medium	System

4.3 Agent Assignment Module

Req ID	Description	Priority	Role
FR-ASN-01	AssignmentEngine scheduler runs every 30 minutes. It finds all SUBMITTED declarations with no active assignment, matches an available agent by PIN code proximity, creates an AgentAssignment record, sets deadline to now + 72 hours, and updates status to UNDER_VERIFICATION.	High	System
FR-ASN-02	Admin can manually assign any available agent to any SUBMITTED declaration. System creates assignment, sets assignedBy to ADMIN, and sends email to agent.	High	Admin
FR-ASN-03	Admin can reassign a declaration from one agent to another. Old assignment is marked REASSIGNED. New assignment is created. Both agents receive email notifications.	High	Admin
FR-ASN-04	Admin can cancel an active assignment. Assignment is marked CANCELLED. Declaration status reverts to SUBMITTED. Agent receives cancellation email.	High	Admin
FR-ASN-05	DeadlineMonitor scheduler runs every hour. If an agent misses the 72-hour deadline, it auto-reassigns the declaration and applies a strike to the agent.	High	System
FR-ASN-06	Three strikes on an agent account can result in suspension.	Medium	Admin
FR-ASN-07	assignedBy field stores SYSTEM for scheduler assignments and ADMIN for manual assignments.	Medium	System

4.4 Agent Verification Module

Req ID	Description	Priority	Role
FR-VER-01	Agent views their assigned verification tasks. Each task shows declaration details, deadline, and user information.	High	Agent
FR-VER-02	Agent starts an assignment which logs GPS coordinates at the time of starting.	High	Agent
FR-VER-03	Agent can CONFIRM declared values, MODIFY incorrect values on-site, or REJECT the declaration with a mandatory rejection reason.	High	Agent
FR-VER-04	Agent-corrected values always override user-declared values in the carbon score calculation.	High	System
FR-VER-05	On rejection, agent must provide a text rejection reason. System updates status to REJECTED and notifies the user.	High	Agent
FR-VER-06	Agent receives 24-hour deadline reminder email before their assignment deadline.	Medium	System

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Req ID	Description	Priority	Role
FR-VER-07	Agent can upload and attach documents or photos as part of the verification process.	Medium	Agent

4.5 Carbon Score Calculation Engine

The CarbonScoreService calculates the carbon score in 8 sequential steps immediately after agent verification is submitted:

Step	Name	Description
1	Duplicate Guard	Check no existing score for this declaration. Throw DuplicateScoreException if found.
2	Load Verified Data	Load declaration data applying agent corrections over user-declared values.
3	Energy CO ₂	Electricity CO ₂ = units (kWh) x 0.82 x 12 (annualised).
4	Transport CO ₂	Sum of vehicle km driven + public transport usage converted to CO ₂ equivalent.
5	Lifestyle / Ops CO ₂	Additional emissions from household lifestyle or MSME operations data.
6	Per Capita	Total CO ₂ divided by number of household members or MSME employees.
7	Zone Classification	Household: Champion < 3,000 kg, Improver 3,000–6,000 kg, Defaulter > 6,000 kg. MSME: Champion < 6,000 kg, Improver 6,000–12,000 kg.
8	Recommendations	Generate top 3 personalised recommendations (HIGH / MEDIUM / LOW priority) based on the highest-emission categories.

4.6 Insurance Discount Engine

Req ID	Description	Priority	Role
FR-DIS-01	Zone discount is applied based on carbon zone: Green Champion earns up to 30%, Improver up to 15%, Defaulter earns 0%.	High	System
FR-DIS-02	Improvement bonus is applied if the user reduced their carbon score from the previous year: ≥30% reduction earns +12%, ≥20% earns +8%, ≥10% earns +5%.	High	System
FR-DIS-03	Duration discount is applied for multi-year policies: 2-year policy earns +10%, 1-year earns +5%.	Medium	System
FR-DIS-04	Total combined discount is capped at a maximum of 35%.	High	System
FR-DIS-05	Discount breakdown is shown to the user line by line: Base Premium, Zone Discount, Improvement Bonus, Duration Discount, Final Premium.	High	User
FR-DIS-06	Policies page shows base prices to all users. Users with a verified score see personalised discounted prices. Users without a score see a prompt to complete their declaration.	High	User

4.7 Insurance Policy Module

Req ID	Description	Priority	Role
FR-POL-01	Admin manages the insurance policy catalog in the backend. Policies are fetched dynamically — no static policy data on the frontend.	High	Admin
FR-POL-02	User browses available policies. If they have a verified carbon score, personalised discount is shown. Otherwise, base prices are shown with a message to complete declaration.	High	User
FR-POL-03	User can purchase a policy. Backend validates the purchase, saves a UserPolicy record, and links it to the user account.	High	User
FR-POL-04	User can view all their purchased policies in My Policies section.	High	User
FR-POL-05	System throws PolicyAlreadyPurchasedException if user tries to buy an already active policy of the same type.	Medium	System

4.8 Notification Module

Req ID	Description	Priority	Role
FR-NOT-01	System creates in-app notifications for all key events: declaration submitted, agent assigned, verification complete, score generated, policy purchased.	High	System
FR-NOT-02	User can mark individual notifications as read by clicking them. Read notifications are removed from the unread list.	High	User
FR-NOT-03	User can click Mark All as Read to clear all unread notifications at once.	High	User
FR-NOT-04	Read status is persisted in the database. Refreshing the page does not bring back read notifications.	High	System
FR-NOT-05	System sends email notifications for all key events via SMTP through EmailService. All email templates are HTML-formatted with GreenSure branding.	High	System
FR-NOT-06	Toast notifications appear in the top-right corner for all user actions. Toasts are colour-coded: green for success, red for error, yellow for warning. Auto-dismiss after 3 seconds.	Medium	All

5. Non-Functional Requirements

5.1 Security

- All API endpoints except /auth/login, /auth/register, /auth/forgot-password, /auth/reset-password are protected by JWT authentication
- Passwords are hashed using BCrypt before storage — plain text passwords are never stored
- JWT tokens are signed with a secret key using JJWT and validated on every request
- Role-based access is enforced at the method level using @PreAuthorize — not just at the URL level
- Password reset tokens expire after 15 minutes
- CSRF is disabled as the API is fully stateless

5.2 Performance

- All admin and agent list views must implement pagination — no unbounded data loads
- Angular routes must be lazy-loaded to minimise initial bundle size
- Scheduler tasks must be lightweight and complete within their interval window
- Email sending must be asynchronous — it must not block the main request thread

5.3 Usability

- Form validation must show red borders on invalid fields and auto-scroll to the first error
- Page must scroll to warning messages when validation errors appear
- Toast notifications must provide immediate feedback for every user action
- Loading states must be shown during all API calls with a consistent loading animation
- All error messages must be human-readable — no technical exception stack traces shown to users

5.4 Maintainability

- Frontend components must be placed inside the components folder — not inside feature folders
- All services must be in the core/services folder following consistent naming
- Backend must follow Controller → Service → Repository layered architecture strictly
- All exceptions must be custom named exceptions handled by GlobalExceptionHandler
- No generic Internal Server Error responses — every error must have a specific custom exception

5.5 Scalability

- H2 in-memory database is used for development. The data layer must be PostgreSQL-ready with a single configuration change
- The automatic agent assignment system scales with the number of declarations and agents without any code changes
- Email sending is decoupled from business logic through a single EmailService class

6. Custom Exception Specification

All exceptions must be handled by the GlobalExceptionHandler using @RestControllerAdvice. Every error response must use the standard ApiResponse<T> format with a timestamp, HTTP status code, and human-readable message. No generic Internal Server Error responses are permitted.

Exception Class	HTTP Status	When Thrown
DeclarationNotFoundException	404	Declaration ID does not exist
DeclarationNotEditableException	400	Attempt to edit a non-editable declaration
DuplicateDeclarationException	409	User tries to submit a second declaration for the same year
MaxResubmissionLimitExceededException	400	User exceeds 3 resubmission attempts on a rejected declaration
DeclarationAlreadyAssignedException	409	Admin tries to assign an already-assigned declaration
AssignmentNotFoundException	404	Assignment ID does not exist
AgentNotFoundException	404	Agent ID does not exist
AgentNotAvailableException	400	Selected agent is not available for assignment
UserNotFoundException	404	User ID or email does not exist
PolicyNotFoundException	404	Policy ID does not exist
PolicyAlreadyPurchasedException	409	User tries to buy an already active policy
InvalidTokenException	400	JWT or password reset token is invalid or expired
AccountSuspendedException	403	Suspended user tries to login
CarbonScoreNotFoundException	404	Score not found for user or declaration
DuplicateScoreException	409	Score already exists for this declaration

7. API Endpoint Summary

7.1 Auth Endpoints

Method	Endpoint	Access	Description
POST	/auth/register	Public	Register new user (HH or MSME)
POST	/auth/login	Public	Login and receive JWT token
POST	/auth/forgot-password	Public	Request password reset email
POST	/auth/reset-password	Public	Reset password using token

7.2 Declaration Endpoints

Method	Endpoint	Access	Description
POST	/declaration/submit	USER	Submit new carbon declaration
PUT	/declaration/{id}/edit	USER	Edit draft or rejected declaration
GET	/declaration/my	USER	Get all declarations for current user
GET	/declaration/{id}	USER	Get single declaration details
GET	/declaration/history	USER	Declaration history with status
GET	/admin/declarations	ADMIN	All declarations platform-wide
GET	/admin/declarations/unassigned	ADMIN	Declarations with no agent assigned
GET	/admin/declarations/assigned	ADMIN	Declarations with active assignment

7.3 Assignment Endpoints

Method	Endpoint	Access	Description
GET	/admin/agents/available	ADMIN	List agents available for assignment
POST	/admin/assignment/assign	ADMIN	Manually assign agent to declaration
PUT	/admin/assignment/reassign	ADMIN	Reassign declaration to different agent
PUT	/admin/assignment/change-agent	ADMIN	Change agent by assignment ID
DELETE	/admin/assignment/cancel/{id}	ADMIN	Cancel active assignment

8. Enhancements & Future Scope

8.1 Pending Enhancements (Next Sprint)

Req ID	Description	Priority	Role
ENH-01	Improve carbon score formula with more accurate CO ₂ coefficients and additional input fields aligned to real carbon standards.	High	System
ENH-02	Implement pagination on all admin and agent list views.	High	Admin
ENH-03	Fix frontend file structure — all components inside components folder, all services in core/services.	Medium	Dev
ENH-04	Add file upload for supporting documents (electricity bills, vehicle RC, solar proof). Agent can verify uploaded documents.	High	Agent
ENH-05	Implement complete insurance flow — user browses, buys policy, agent acts as underwriter, claims officer handles claims.	High	User
ENH-06	Research and align carbon score methodology with real-world green insurance and net zero standards.	Medium	Dev

8.2 Future Scope — Roadmap

Next 3 Months — Production Ready

- Replace H2 with PostgreSQL for production database
- Deploy backend on AWS or Railway
- Deploy Angular frontend on Vercel or Netlify
- Angular HTTP interceptors to remove manual JWT header injection
- Full custom exception hierarchy across all services

Next 6 Months — Scale & Compliance

- Mobile app using Angular and Capacitor
- Real IRDAI-compliant insurance policy integration
- Google Maps API for agent proximity matching instead of PIN code
- Carbon score public API for third-party platform integration
- Bulk CSV declaration upload for large MSMEs
- Admin live carbon heatmap with real chart visualisation

Long Term Vision — Intelligence & Ecosystem

- ML model for carbon declaration anomaly detection before agent visit
- IoT integration — smart meter auto-populates electricity data in declaration
- Carbon credit marketplace — Champion users earn tradeable carbon credits
- Commercial vehicle fleet insurance expansion
- Open banking integration — utility bills auto-imported into declaration
- National carbon registry integration

9. Core Data Model

The following 12 core entities form the GreenSure data model:

Entity	Key Fields & Purpose
User	id, name, email, password (BCrypt), role, userType, isActive, resubmission_count, passwordResetToken, passwordResetTokenExpiry
HouseholdProfile	id, user_id, numberOfMembers, address, pinCode, city, state
MsmeProfile	id, user_id, businessName, numberOfEmployees, businessType, address, pinCode
CarbonDeclaration	id, user_id, declarationYear, status (DRAFT/SUBMITTED/UNDER_VERIFICATION/VERIFIED/REJECTED), resubmissionCount, submittedAt
DeclarationVehicle	id, declaration_id, vehicleType, fuelType, kmPerYear, vehicleCount
AgentAssignment	id, declaration_id, agent_id, assignedBy (SYSTEM/ADMIN), assignmentStatus (ACTIVE/COMPLETED/REASSIGNED/CANCELLED), deadline, reassignReason
Verification	id, assignment_id, agent_id, outcome (CONFIRMED/MODIFIED/REJECTED), rejectionReason, gpsLat, gpsLng, verifiedAt
CarbonScore	id, user_id, declaration_id, totalCo2, energyCo2, transportCo2, lifestyleCo2, perCapitaCo2, zone, discountPercent, generatedAt
Policy	id, name, description, basePrice, policyType, isActive
UserPolicy	id, user_id, policy_id, purchasedAt, discountApplied, finalPrice, status
Recommendation	id, user_id, declaration_id, category, priority (HIGH/MEDIUM/LOW), description, generatedAt
Notification	id, user_id, message, isRead, createdAt, notificationType